

MATHEMATICS

TARGET : JEE- Advanced

CAPS-8**COMPLEX NUMBER****SCQ (Single Correct Type) :**

- The least value of $|z - 3 - 4i|^2 + |z + 2 - 7i|^2 + |z - 5 + 2i|^2$ occurs when
(A) $1 + 3i$ (B) $3 + 3i$ (C) $3 + 4i$ (D) none of these
- $|z - 3i + 4| + |z - ki + 4| = k$
(A) 0 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) 2
- If $|z|^2 + \bar{A}z^2 + A\bar{z}^2 + B\bar{z} + \bar{B}z + c = 0$ represents a pair of intersecting lines with angle of intersection θ then the value of $|A|$ is
(A) $\tan \theta$ (B) $\cos \theta$ (C) $\sec \theta$ (D) $\frac{\sec \theta}{2}$
- If $|z - z_1| = |z_1|$ and $|z - z_2| = |z_2|$ be the of two circles if the two circles touch each other then
(A) $\operatorname{Re}(z_1 z_2) = 0$ (B) $\operatorname{Re}\left(\frac{z_1}{z_2}\right) = 0$ (C) $I_m(z_1 z_2) = 0$ (D) $I_m\left(\frac{z_1}{z_2}\right) = 0$
- If $|z_1| = 2$, $|z_2| = 3$, $|z_3| = 4$ and $|2z_1 + 3z_2 + 4z_3| = 4$, then absolute value of $8z_2 z_3 + 27z_3 z_1 + 64z_1 z_2$ equals
(A) 24 (B) 48 (C) 72 (D) 96
- If $z = x + iy$ then the equation of a straight line $Ax + By + C = 0$ where $A, B, C \in \mathbb{R}$, can be written on the complex plane in the form $\bar{a}z + a\bar{z} + 2C = 0$ where 'a' is equal to :
(A) $\frac{(A + iB)}{2}$ (B) $\frac{A - iB}{2}$ (C) $A + iB$ (D) none

MCQ (One or more than one correct) :

- If from a point P, representing the complex number z_1 , on the curve $|z| = 2$, two tangents are drawn to the curve $|z| = 1$, meeting the curve at points $Q(z_2)$ and $Q(z_3)$, then _____.
(A) the complex number $\frac{z_1 + z_2 + z_3}{3}$ will lie on the curve $|z| = 1$
(B) $\left(\frac{4}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right)\left(\frac{4}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3}\right) = 9$
(C) $\arg\left(\frac{z_2}{z_3}\right) = \frac{2\pi}{3}$
(D) orthocenter and circumcenter of ΔPQR will coincide

8. If z_1, z_2, z_3 are any three roots of the equation $z^6 = (z+1)^6$, then $\arg \left(\frac{z_1 - z_3}{z_2 - z_3} \right)$ can be equal to _____.
- (A) 0 (B) π (C) $\frac{\pi}{4}$ (D) $-\frac{\pi}{4}$
9. Z_1, Z_2 are two complex numbers satisfying $i|z_1|^2 z_2 - |z_2|^2 z_1 = z_1 - iz_2$. Then which of the following is/are correct.
- (A) $\operatorname{Re} \left(\frac{z_1}{z_2} \right) = 0$ (B) $\operatorname{Im} \left(\frac{z_1}{z_2} \right) = 0$
- (C) $z_1 \bar{z}_2 + \bar{z}_1 z_2 = 0$ (D) $|z_1||z_2| = 1$ or $|z_1| = |z_2|$
10. The curve represented by $z = \frac{3}{2 + \cos \theta + i \sin \theta}$, $\theta \in [0, 2\pi)$
- (A) never meets the imaginary axis (B) meets the real axis in exactly two points
- (C) has maximum value of $|z|$ as 3 (D) has minimum value of $|z|$ as 1

Numerical based Questions :

11. Let z_1, z_2, z_3 be three complex number such that $|z_1| = |z_2| = |z_3| = 1$ and $\frac{z_1^2}{z_2 z_3} + \frac{z_2^2}{z_1 z_3} + \frac{z_3^2}{z_1 z_2} + 1 = 0$, then sum of possible values of $|z_1 + z_2 + z_3|$ is _____.
12. If the vertices of a triangle ABC, A (z_1), B (z_2) and C (z_3) lie on the circle $|z - 3| = 16$, such that $z_1 + z_2 + z_3 = 9$, then the value of $\frac{z_1 \tan A + z_2 \tan B + z_3 \tan C}{\tan A \tan B \tan C}$ is equal to _____.
13. If ω is a non-real complex root of $z^{28} = 1$ and such that $|\omega + 1|$ is maximum and $x = \frac{1}{2} \left| \omega - \frac{1}{\omega} \right|$, then $8x^4 + 4x^3 - 8x^2 - 3x + 4$ is equal to _____.
14. Let z_1, z_2, z_3 be complex numbers such that $z_1 + z_2 + z_3 = 0$ and $|z_1| = |z_2| = |z_3|$ then $z_1^2 + z_2^2 + z_3^2$ is _____.
15. Let $1, \alpha_1, \alpha_2, \alpha_3, \dots, \alpha_{10}$ be the eleven 11th roots of unity. Let $\lambda = \sum_{r=1}^{10} r(\alpha_r + \alpha_{11-r})$. The value of $\frac{\lambda + 11}{11}$ equals _____.
16. If a and b are positive integer such that $N = (a + ib)^3 - 107i$ is a positive integer then find the value of $\frac{N}{2}$

17. Let z, w be complex numbers such that $\bar{z} + i\bar{w} = 0$ and $\arg zw = \pi$. If $\operatorname{Re}(z) < 0$ and principal $\arg z = \frac{a\pi}{b}$ then find the value of $a + b$. (where a & b are co-prime natural numbers)
18. Let z is a complex number satisfying the equation, $z^3 - (3 + i)z + m + 2i = 0$, where $m \in \mathbb{R}$. Suppose the equation has a real root α , then find the value of $\alpha^4 + m^4$
19. If $x = 9^{1/3} 9^{1/9} 9^{1/27} \dots \infty$, $y = 4^{1/3} 4^{-1/9} 4^{1/27} \dots \infty$, and $z = \sum_{r=1}^{\infty} (1+i)^{-r}$ and principal argument of $P = (x + yz)$ is $-\tan^{-1} \left(\frac{\sqrt{a}}{b} \right)$ then determine $a^2 + b^2$. (where a & b are co-prime natural numbers)
20. $z_1, z_2 \in \mathbb{C}$ and $z_1^2 + z_2^2 \in \mathbb{R}$,
 $z_1(z_1^2 - 3z_2^2) = 2$, $z_2(3z_1^2 - z_2^2) = 11$
 If $z_1^2 + z_2^2 = \lambda$ then determine λ^2
21. Let $|z| = 2$ and $w = \frac{z+1}{z-1}$ where $z, w \in \mathbb{C}$ (where $\mathbb{C}$ is the set of complex numbers), then find product of maximum and minimum value of $|w|$.
22. If ω and ω^2 are the non-real cube roots of unity and $a, b, c \in \mathbb{R}$ such that $\frac{1}{a+\omega} + \frac{1}{b+\omega} + \frac{1}{c+\omega} = 2\omega^2$ and $\frac{1}{a+\omega^2} + \frac{1}{b+\omega^2} + \frac{1}{c+\omega^2} = 2\omega$. If $\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1} = \lambda$ then determine λ^4

Matrix Match Type :**23. Match the following:**

Consider the two circles in the complex plane

$$C_1 : \left| \frac{z+i}{z-1} \right| = 2$$

$$C_2 : z\bar{z} - (3+4i)z - (3-4i)\bar{z} + 9 = 0$$

Let (α_1, β_1) be the centre of C_1 and (α_2, β_2) be that of C_2

Column-I

- (A) $2\alpha_1 + \beta_1$ equals
 (B) $\alpha_2 - 2\beta_2$ equals
 (C) The radius of circle C_1 equals
 (D) The radius of circle C_2 equals

Column-II

- (p) 11
 (q) 3
 (r) 4
 (s) $\frac{2\sqrt{2}}{3}$
 (t) 2

Options:

- (A) $A \rightarrow q; B \rightarrow p; C \rightarrow s; D \rightarrow r;$ (B) $A \rightarrow q; B \rightarrow r; C \rightarrow s; D \rightarrow p;$
 (C) $A \rightarrow q; B \rightarrow s; C \rightarrow p; D \rightarrow r;$ (D) $A \rightarrow p; B \rightarrow q; C \rightarrow s; D \rightarrow r;$

Subjective based Questions :

24. If z_1 and z_2 are the two complex numbers satisfying $|z - 3 - 4i| = 8$ and $\text{Arg}\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$ then find the range of the values of $|z_1 - z_2|$.
25. Let a, b, c be distinct complex numbers such that $\frac{a}{1-b} = \frac{b}{1-c} = \frac{c}{1-a} = k$, ($a, b, c \neq 1$). Find the value of k .

ANSWER KEY OF CAPS-8

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|------------------------------|-----------|----------|---|------------|
| 1. (D) | 2. (D) | 3. (D) | 4. (D) | 5. (D) |
| 6. (C) | 7. (ABCD) | 8. (AB) | 9. (ACD) | 10. (ABCD) |
| 11. (3) | 12. (3) | 13. (3) | 14. (0) | 15. (0) |
| 16. (99) | 17. (7) | 18. (32) | 19. (13) | 20. (25) |
| 21. (1) | 22. (16) | 23. (A) | 24. $ (z_1 - z_2) \in [3\sqrt{2}, 13\sqrt{2}]$ | |
| 25. $-\omega$ or $-\omega^2$ | | | | |